

MH370 broad-flight uncertainty through 00:11 — numerical diagnostic

Broad multi-mode, repeated-event filtering diagnostic

Numerical support: FAILED

No truth loaded; no accuracy claims

Scientific families (never pooled)

- BTO only: 8 deterministic seed(s)
- BTO + BFO (7 Hz): 8 deterministic seed(s)

Declared control-history support

Five initial modes (declared prior):
CTH 20.0% CMH 20.0% CTT 20.0%
CMT 20.0% GC 20.0%

Four recurring-event rates (declared prior):
tau=640s 25.0% tau=2024s 25.0%
tau=6402s 25.0% tau=20243s 25.0%

Every rate family enables recurring lateral, speed, and altitude renewal streams. Each lateral renewal can reselect any of the five modes and draw across the declared ± 180 -degree course-change support; speed and altitude commands renew independently. All displayed prior weights are declared and uncalibrated.

Interpretation limits

- No assumption-free or empirically calibrated true flight-path probability is claimed.
- Control modes, event-rate families, event marks, performance bounds, and their prior probabilities are declared model choices.
- Top-level scientific families are not pooled; seed pooling is numerical only and equal within a family.
- One known-flight trajectory cannot calibrate general coverage of the accident-flight technique.
- Map density is display-smoothed; all metrics use unsmoothed particle weights.
- A configured fuel-state anchor is shown as a state reset with no positional likelihood; it is not a SATCOM measurement or a truth-scored epoch.

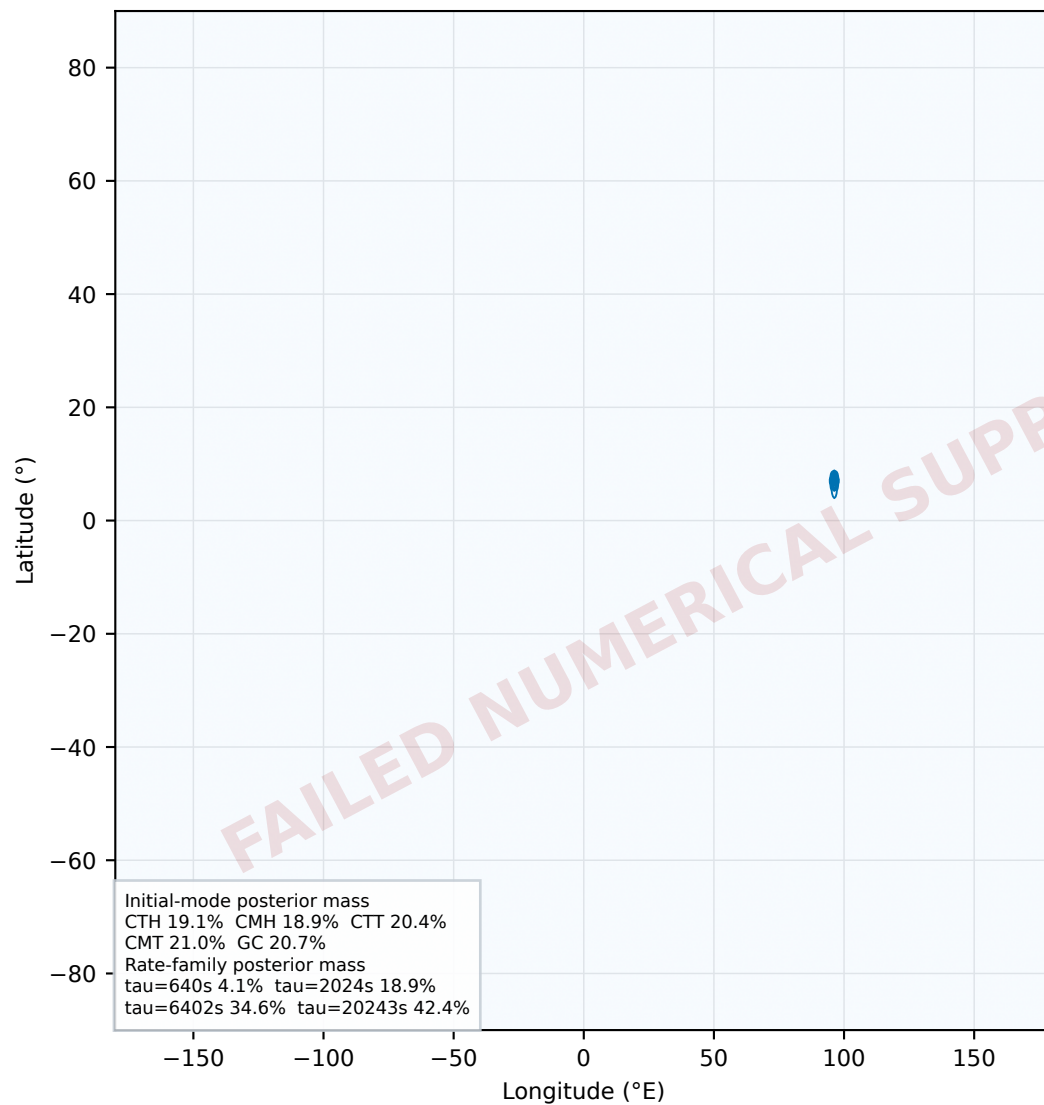
Failed/unassessed criteria: 1893 of 4687

Complete per-epoch criterion ledger: companion JSON audit.

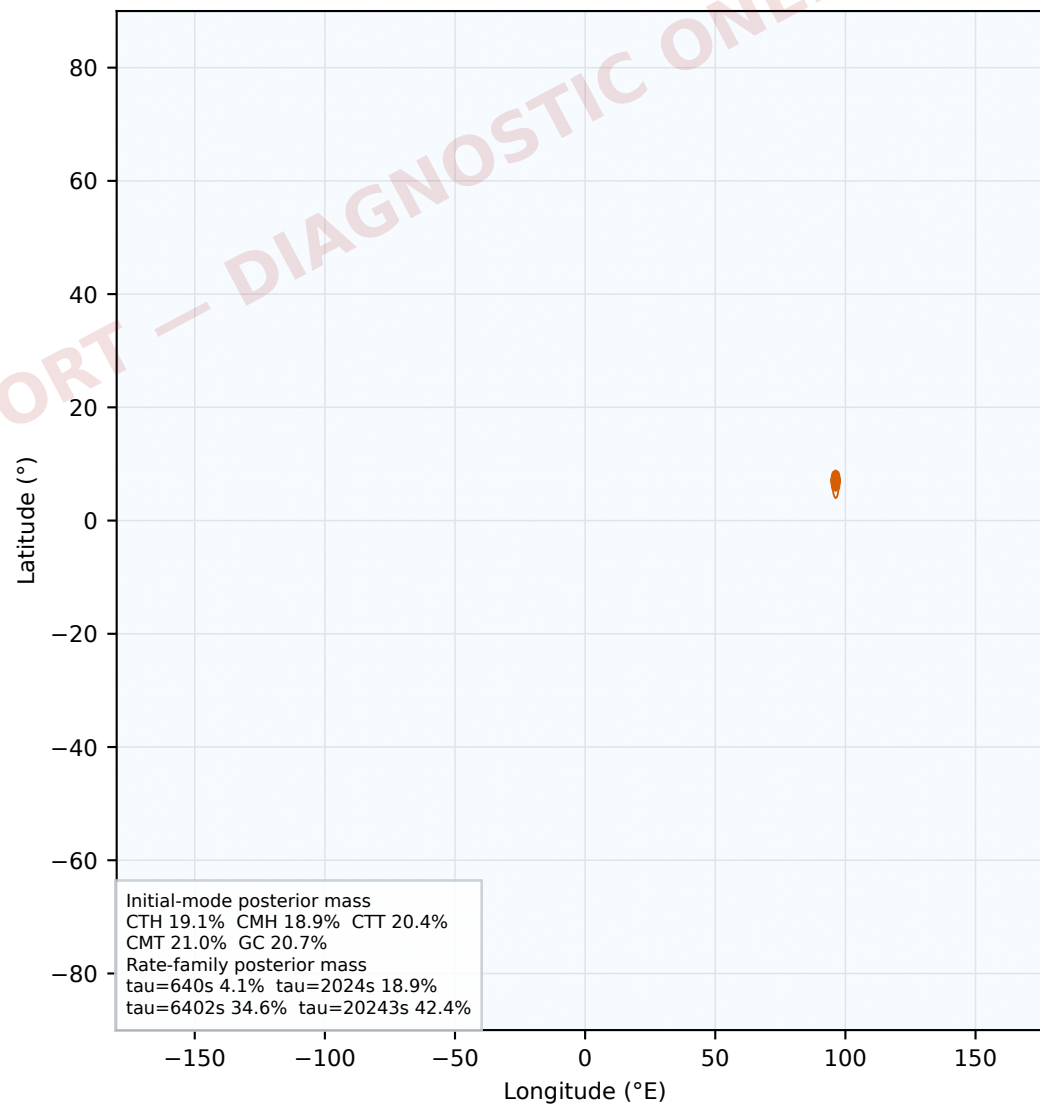
Every spatial probability shown is conditional on the declared model and evidence family. It is not a calibrated 'true uncertainty'.

MH370 — m1825 filtering posterior

BTO only (kept separate)
SATCOM likelihood update: BTO
particle/root ESS 44573 / 44573.1
mean lateral/speed/altitude events 0.17 / 0.18 / 0.26
mass with ≥ 2 lateral events 2.9%
seed TV max 0.017

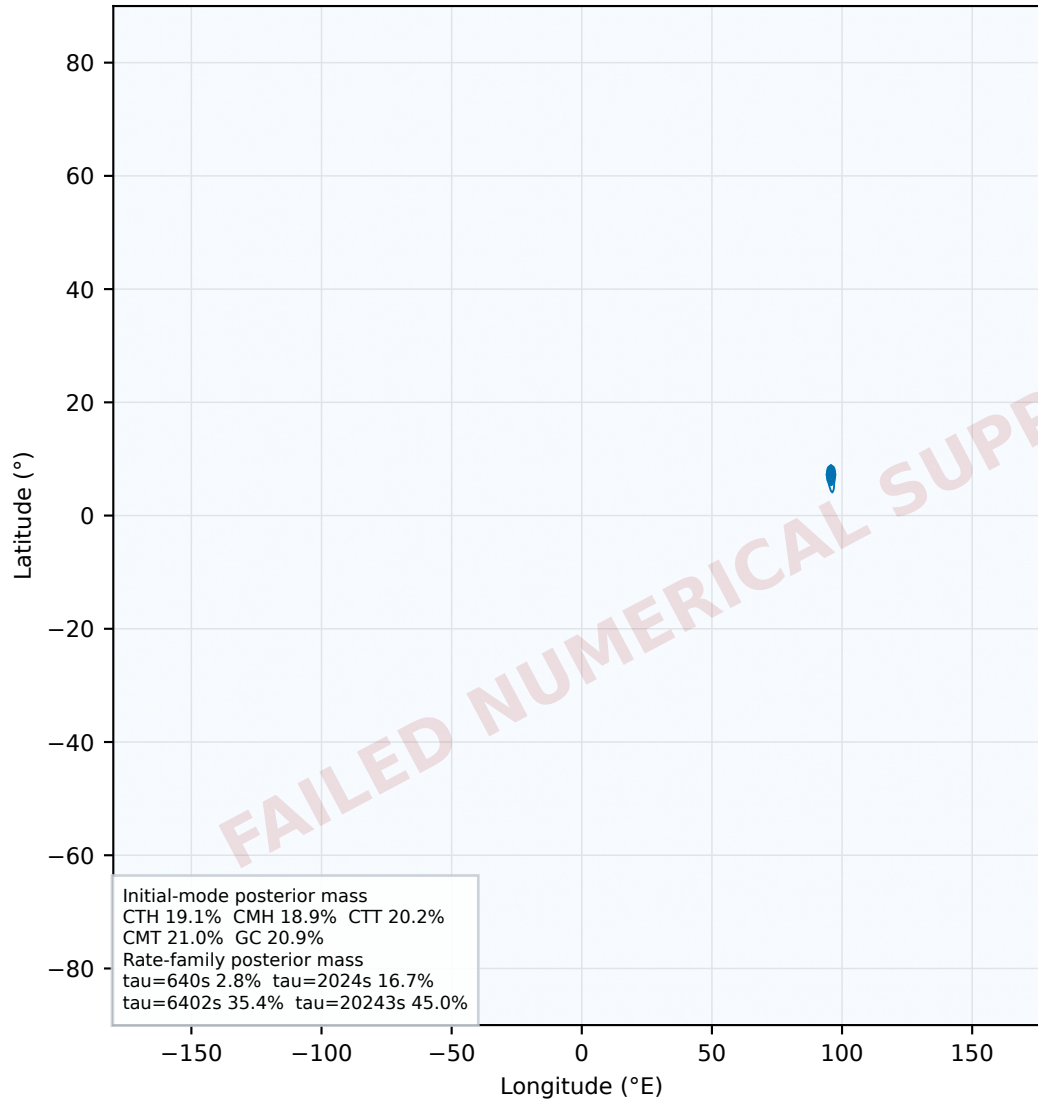


BTO + BFO (7 Hz) (kept separate)
SATCOM likelihood update: BTO
particle/root ESS 44573 / 44573.1
mean lateral/speed/altitude events 0.17 / 0.18 / 0.26
mass with ≥ 2 lateral events 2.9%
seed TV max 0.017

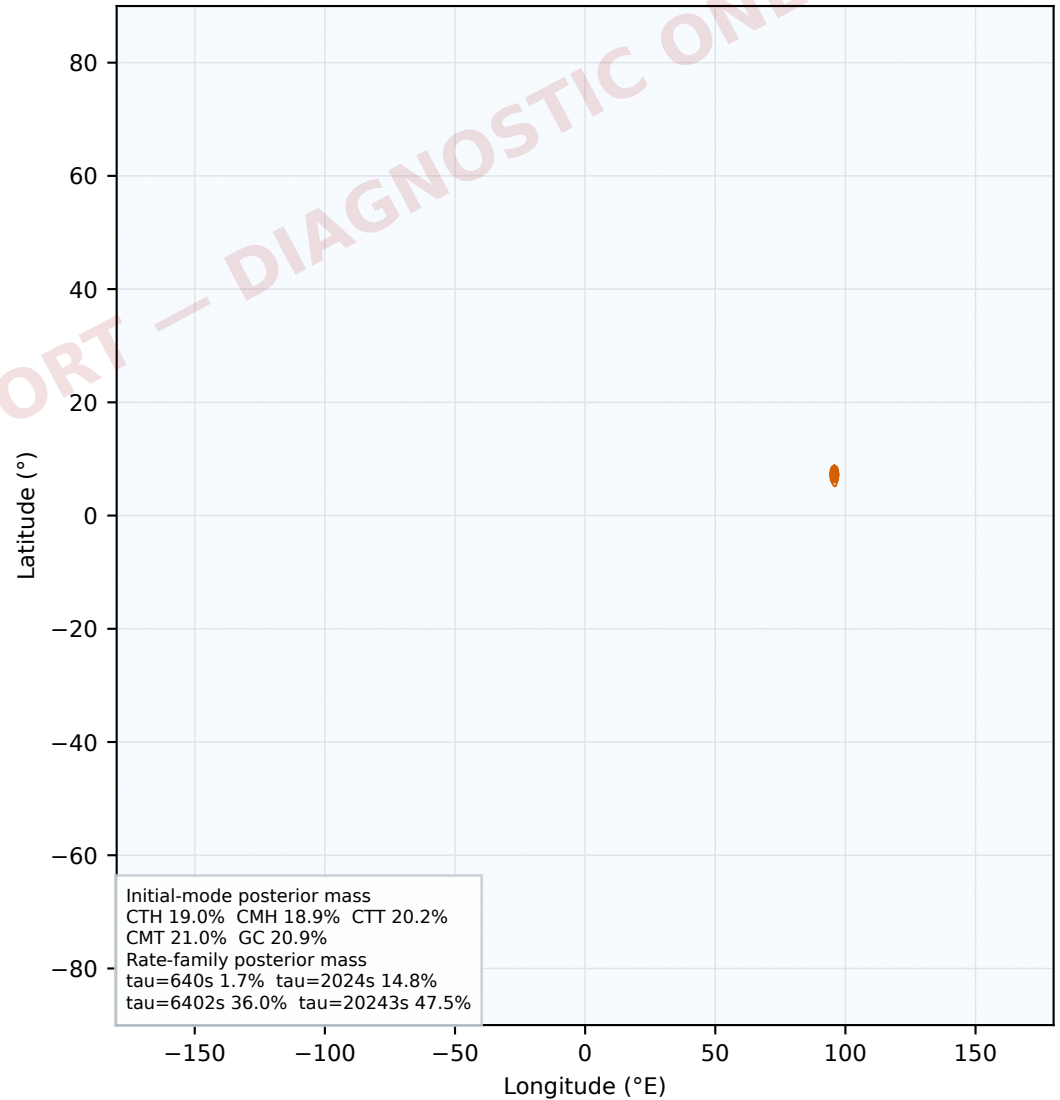


MH370 — m1828a filtering posterior

BTO only (kept separate)
SATCOM likelihood update: BTO
particle/root ESS 61532 / 27826.4
mean lateral/speed/altitude events 0.14 / 0.17 / 0.24
mass with ≥ 2 lateral events 2.3%
seed TV max 0.012

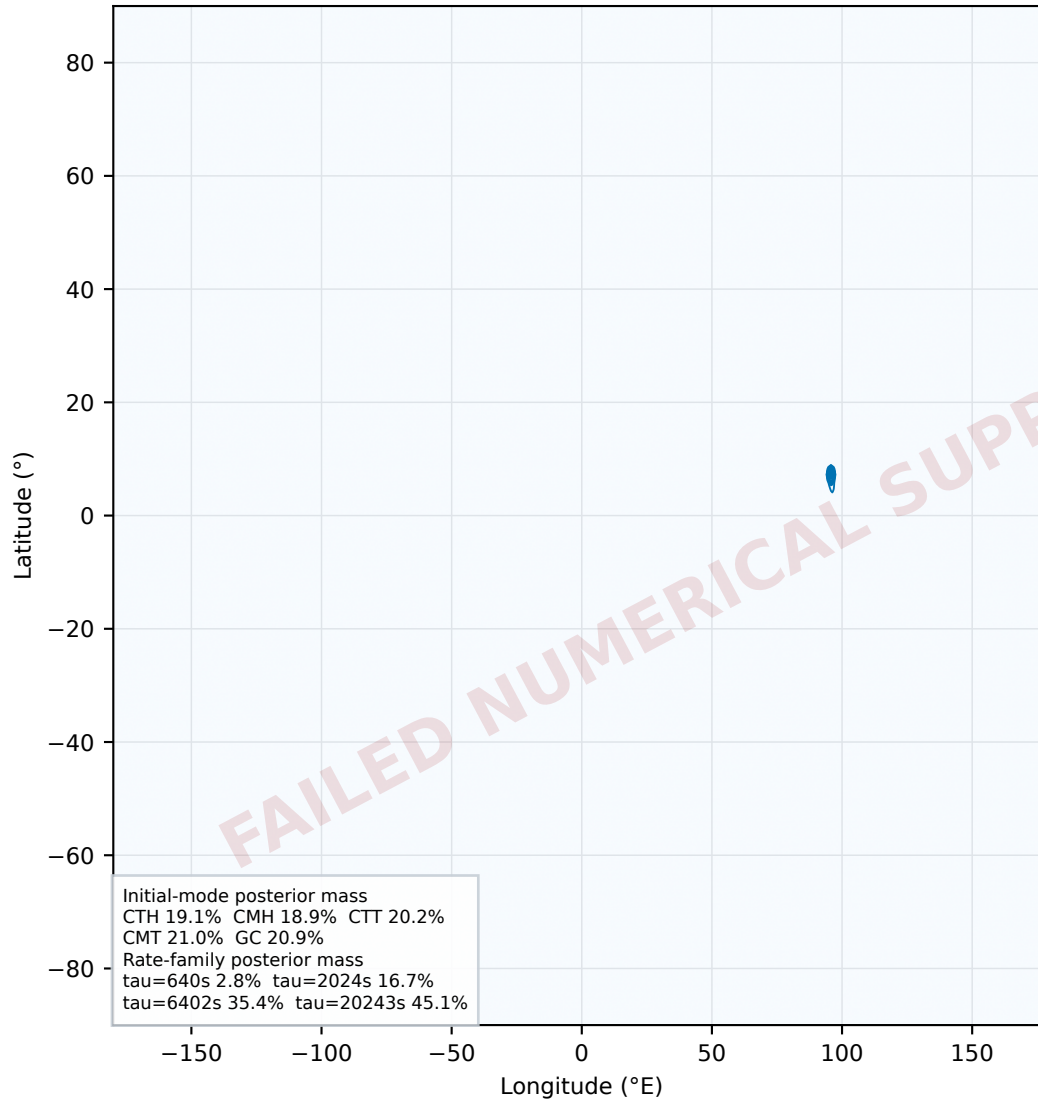


BTO + BFO (7 Hz) (kept separate)
SATCOM likelihood update: BTO + BFO
particle/root ESS 56652 / 25888.3
mean lateral/speed/altitude events 0.08 / 0.14 / 0.19
mass with ≥ 2 lateral events 1.3%
seed TV max 0.007

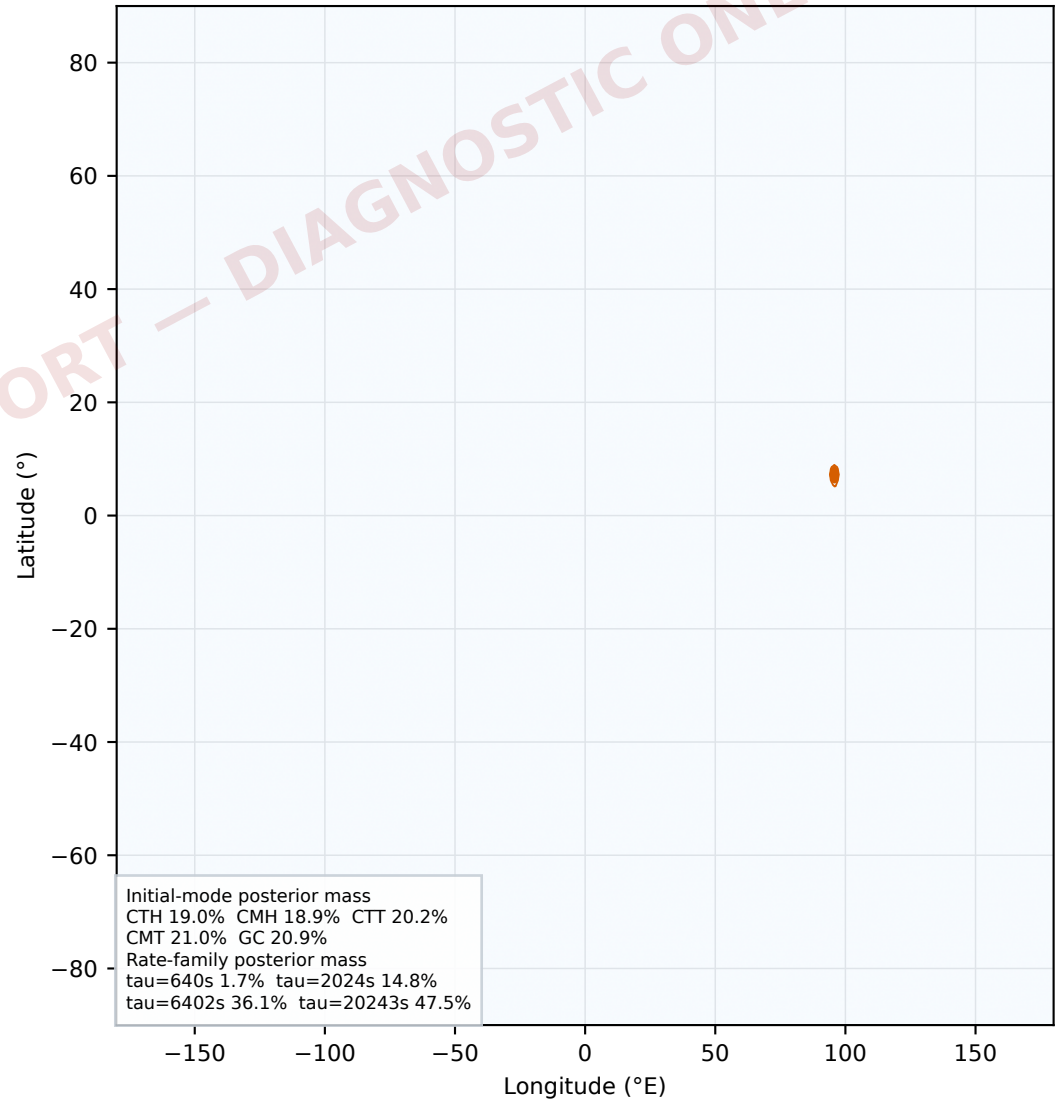


MH370 — arc1-official-calculated-fuel-anchor filtering posterior

BTO only (kept separate)
fuel-state anchor/reset; no positional likelihood
particle/root ESS 80094 / 27213.4
mean lateral/speed/altitude events 0.14 / 0.17 / 0.24
mass with ≥ 2 lateral events 2.4%
seed TV max 0.013

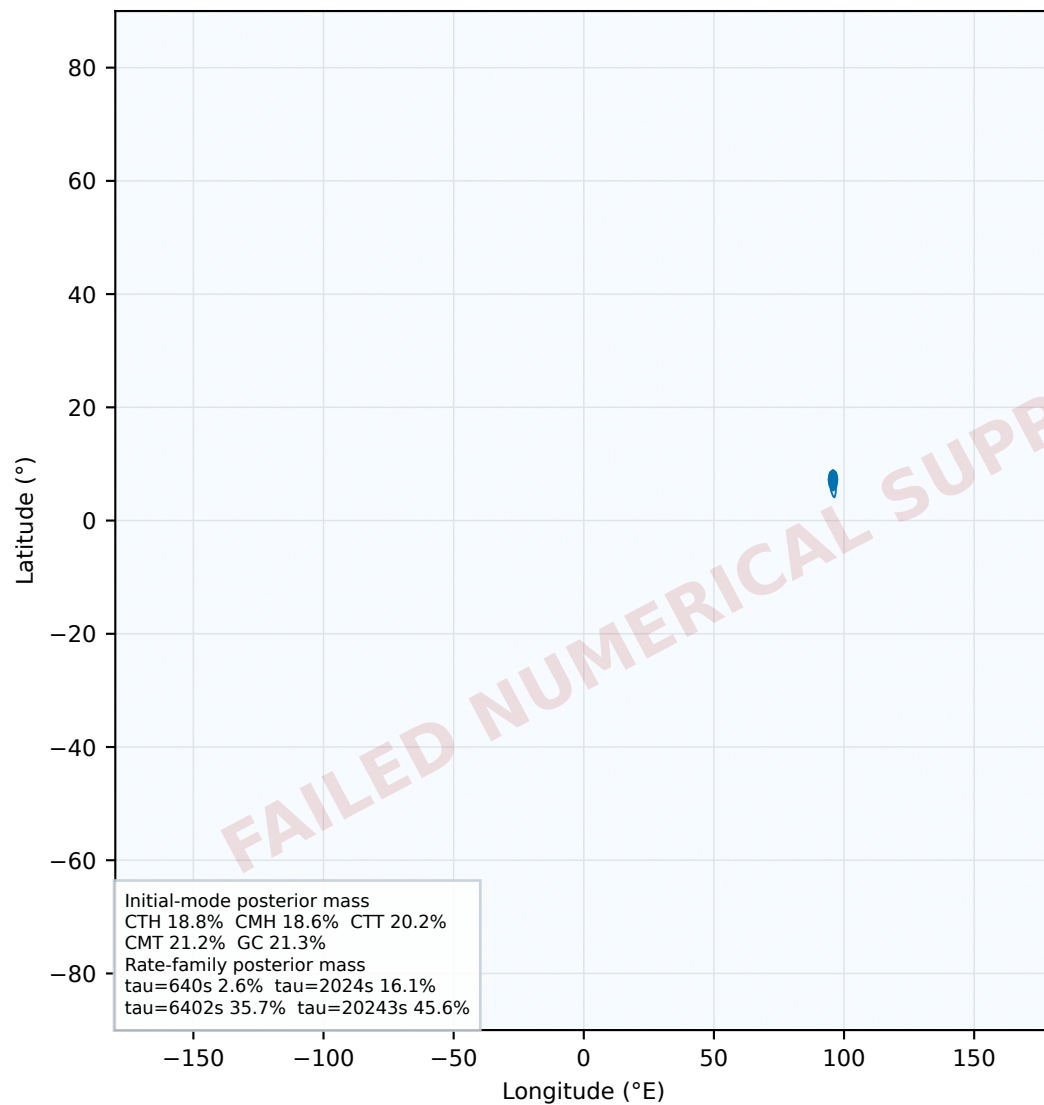


BTO + BFO (7 Hz) (kept separate)
fuel-state anchor/reset; no positional likelihood
particle/root ESS 75419 / 25334.8
mean lateral/speed/altitude events 0.08 / 0.14 / 0.19
mass with ≥ 2 lateral events 1.3%
seed TV max 0.007

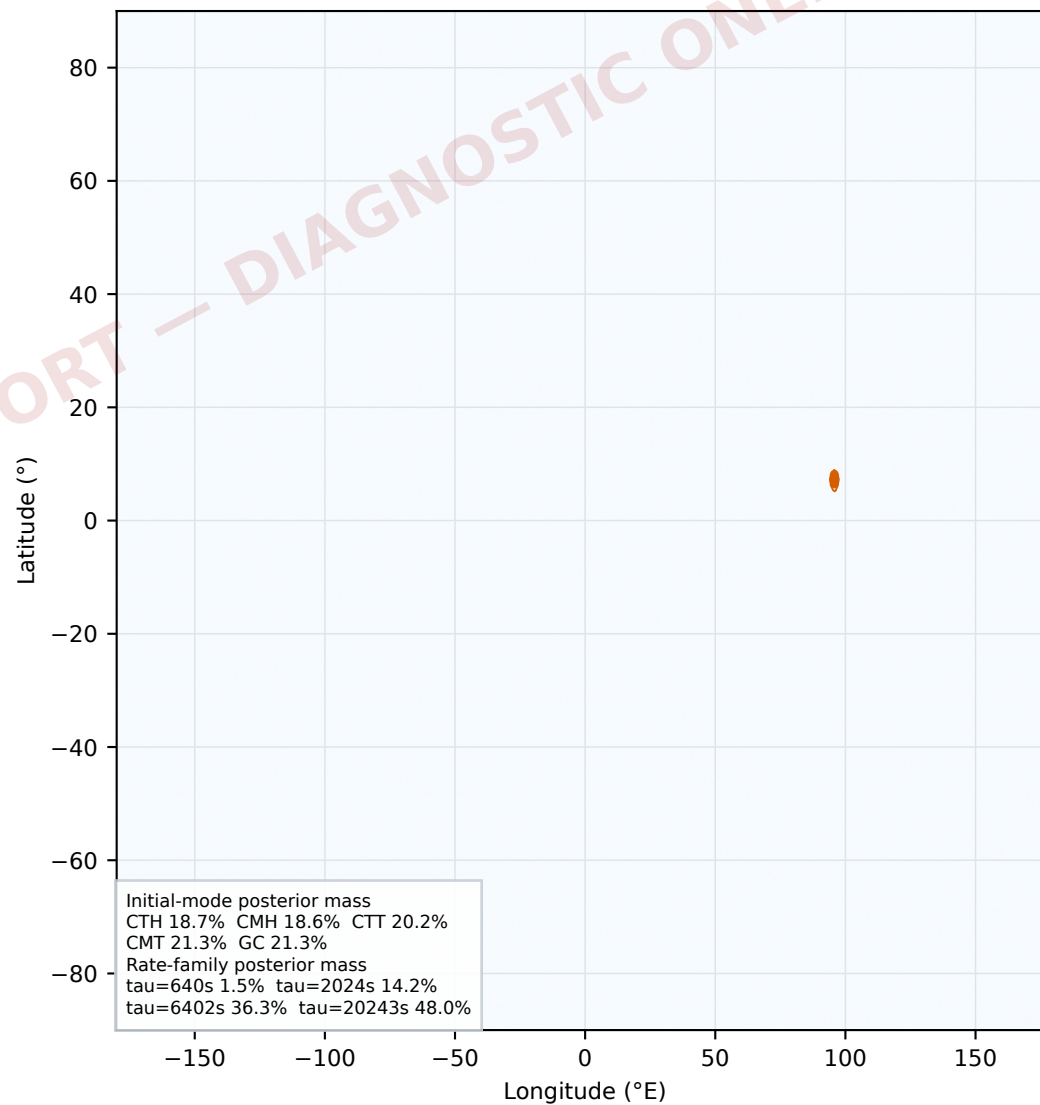


MH370 — m1828b filtering posterior

BTO only (kept separate)
SATCOM likelihood update: BTO
particle/root ESS 66299 / 20611.7
mean lateral/speed/altitude events 0.13 / 0.16 / 0.23
mass with ≥ 2 lateral events 2.3%
seed TV max 0.015

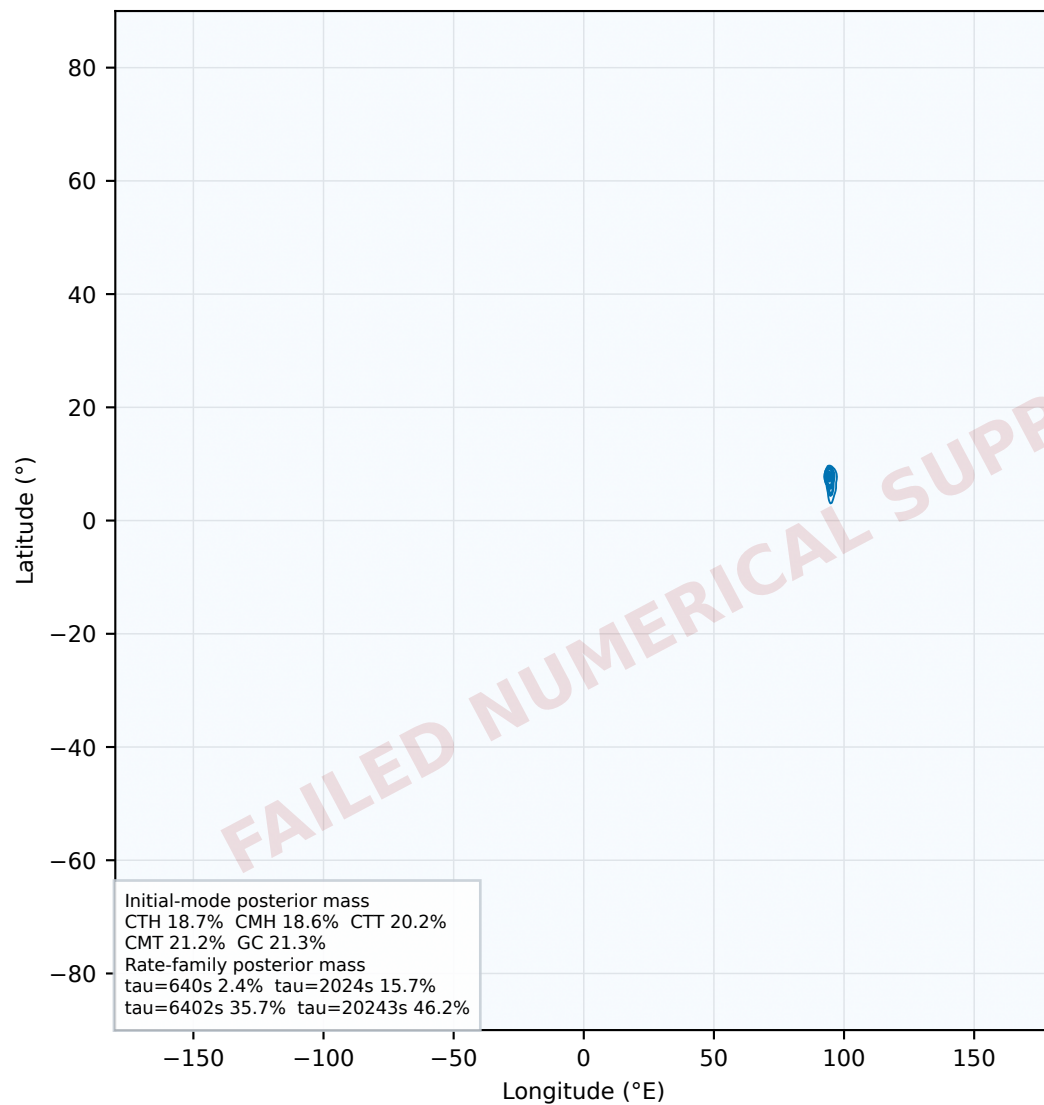


BTO + BFO (7 Hz) (kept separate)
SATCOM likelihood update: BTO + BFO
particle/root ESS 64178 / 19087.3
mean lateral/speed/altitude events 0.07 / 0.13 / 0.18
mass with ≥ 2 lateral events 1.2%
seed TV max 0.009

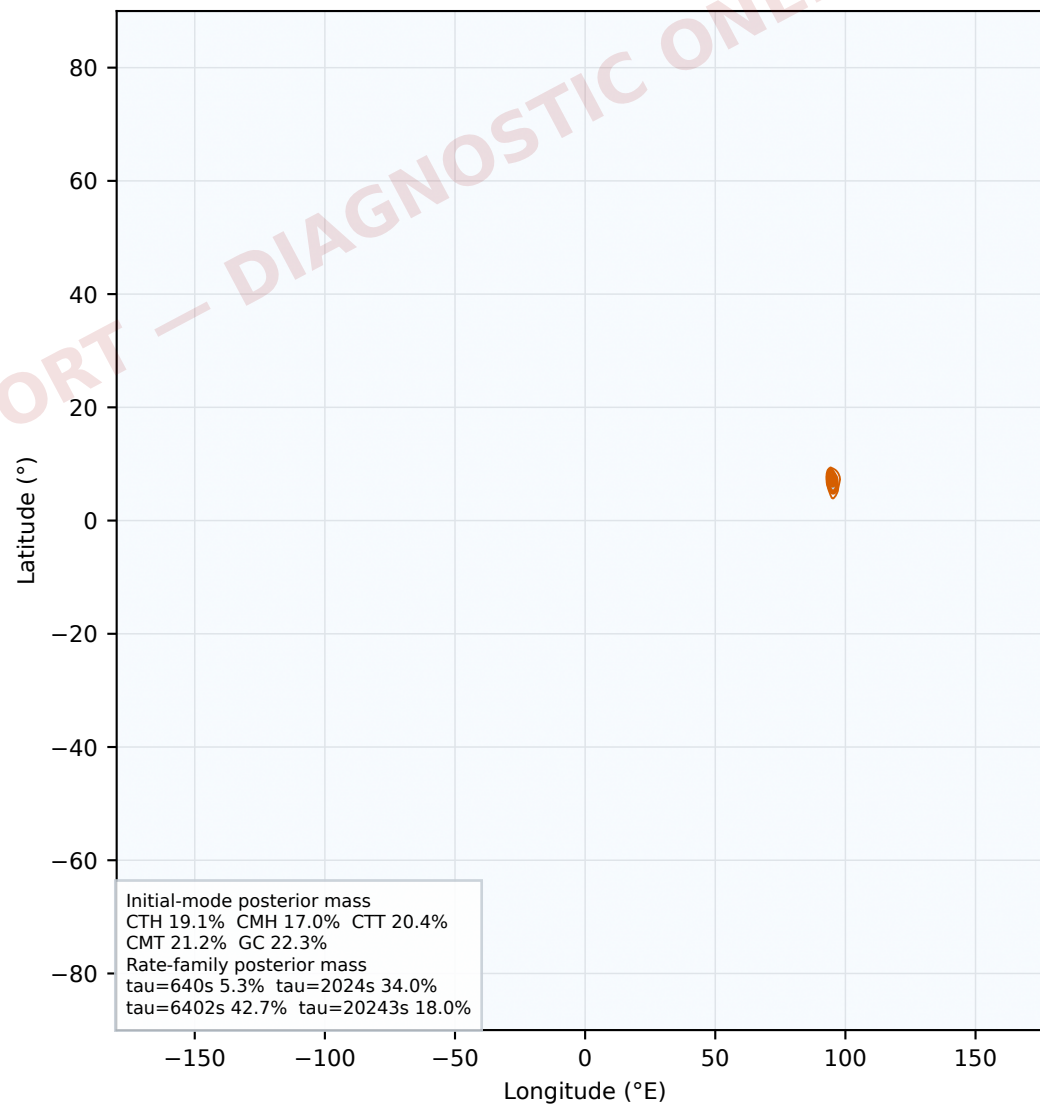


MH370 — m1839 filtering posterior

BTO only (kept separate)
propagation-only checkpoint; likelihood disabled
particle/root ESS 75423 / 20141.9
mean lateral/speed/altitude events 0.27 / 0.29 / 0.35
mass with ≥ 2 lateral events 4.9%
seed TV max 0.040

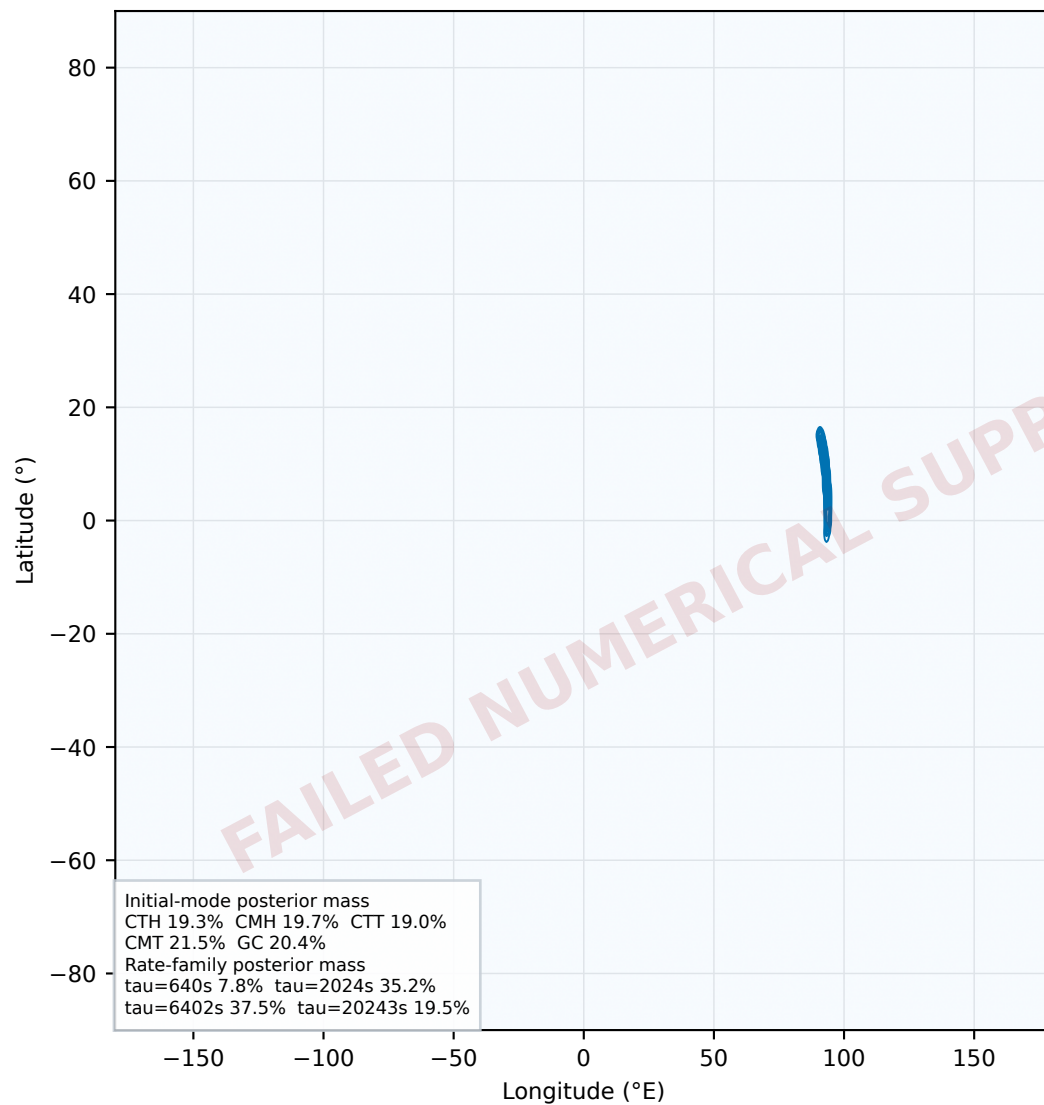


BTO + BFO (7 Hz) (kept separate)
SATCOM likelihood update: BFO
particle/root ESS 4118 / 1630.3
mean lateral/speed/altitude events 1.05 / 0.44 / 0.69
mass with ≥ 2 lateral events 14.2%
seed TV max 0.170

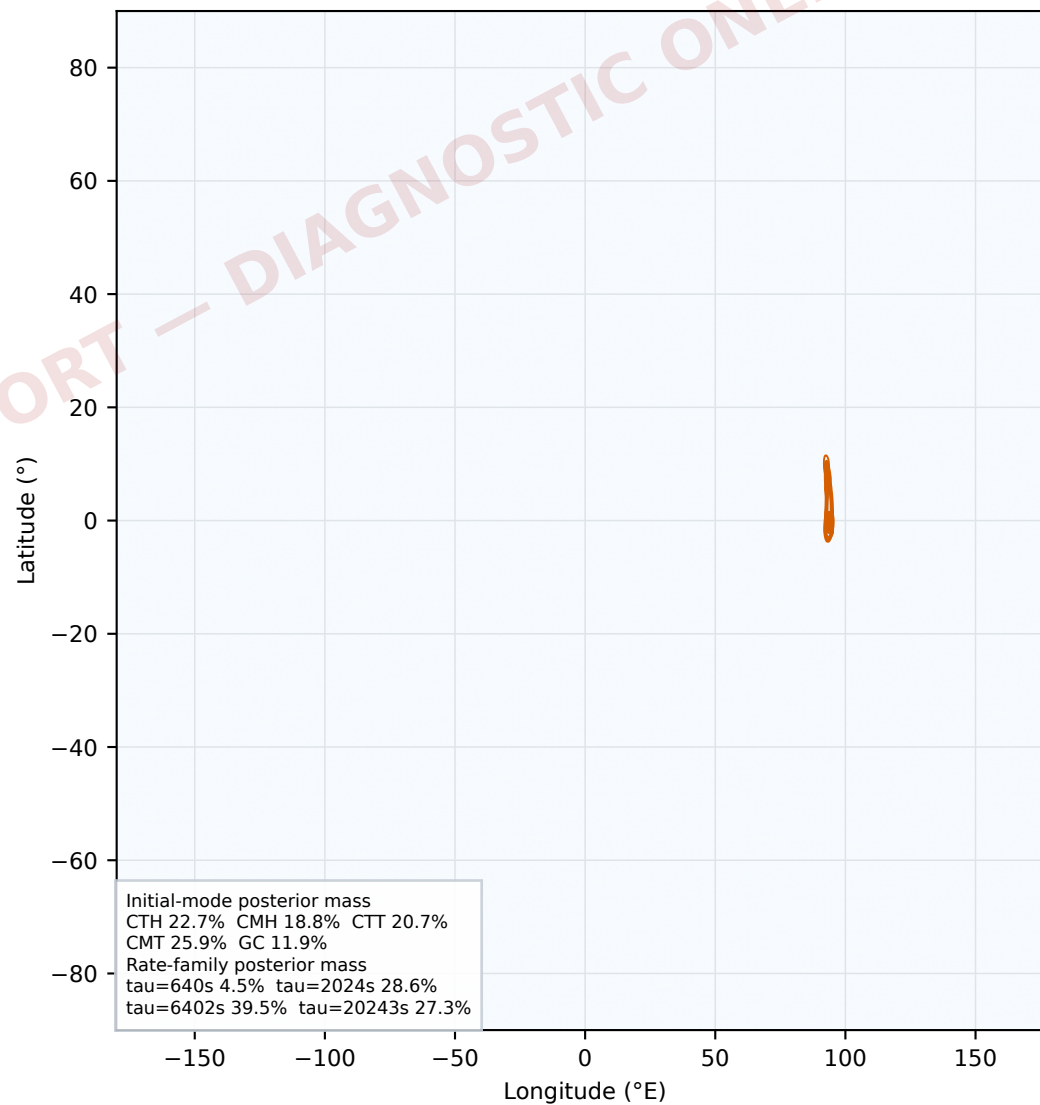


MH370 — m1941 filtering posterior

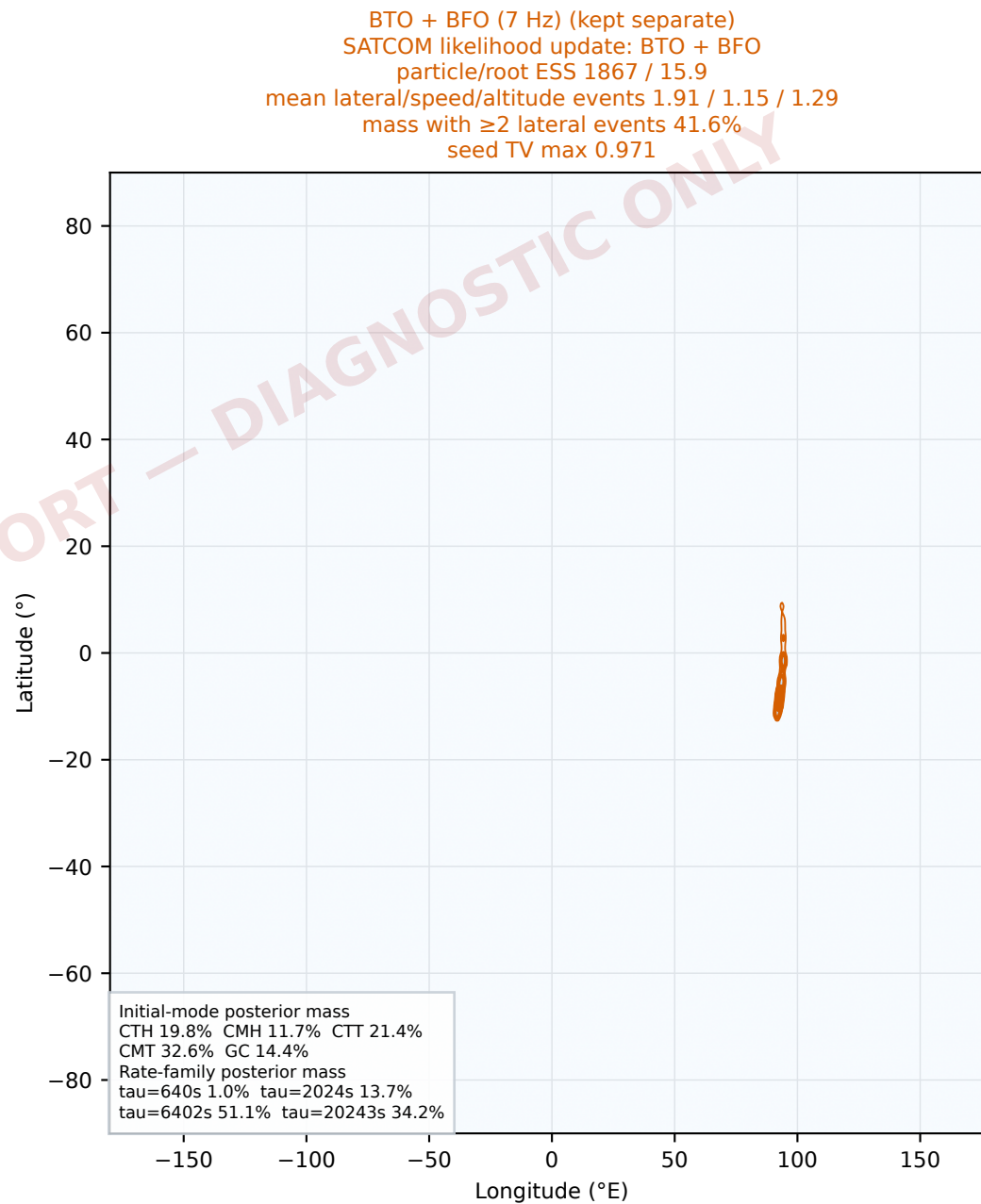
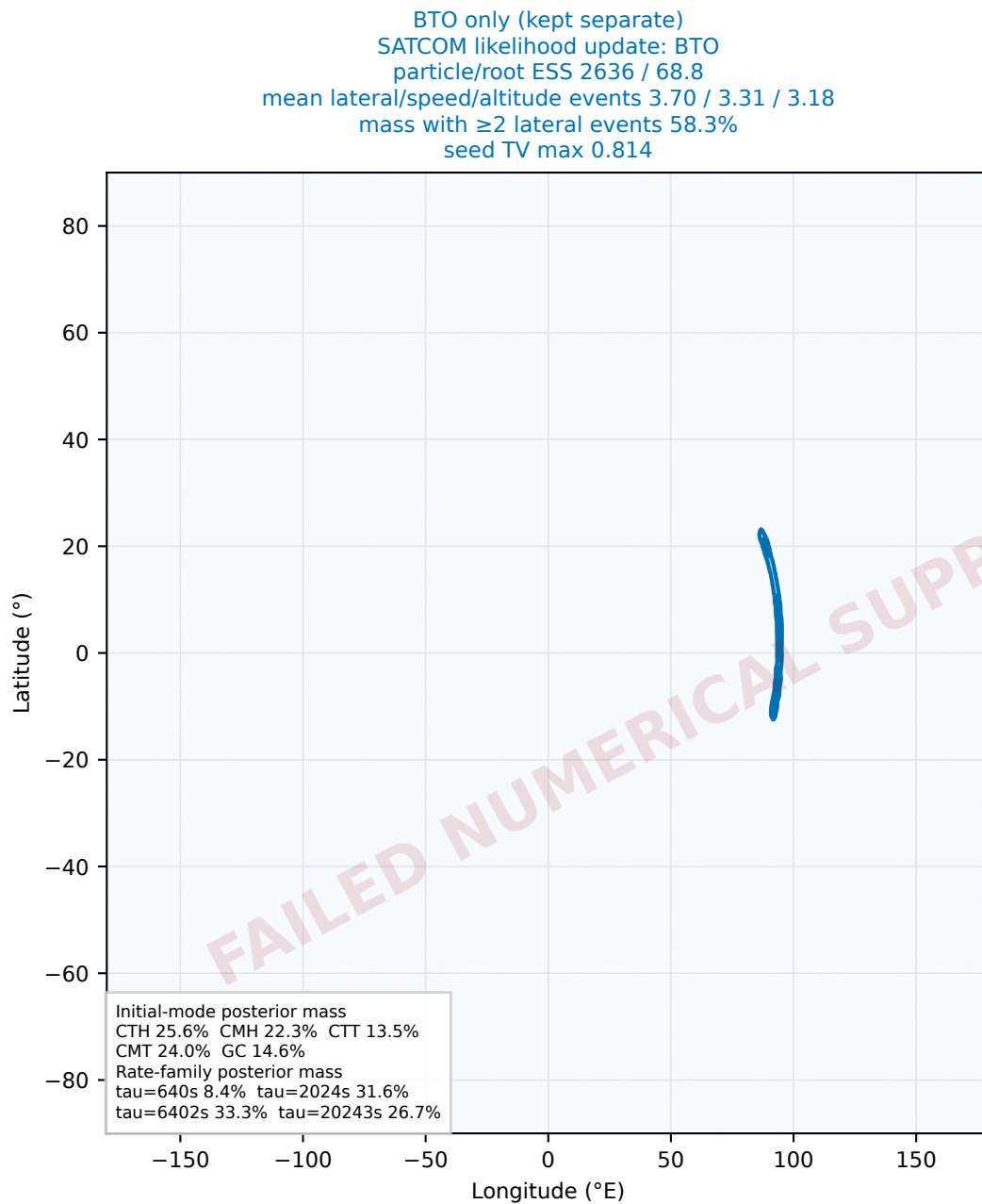
BTO only (kept separate)
SATCOM likelihood update: BTO
particle/root ESS 1338 / 1130.5
mean lateral/speed/altitude events 2.53 / 2.01 / 1.93
mass with ≥ 2 lateral events 57.1%
seed TV max 0.366



BTO + BFO (7 Hz) (kept separate)
SATCOM likelihood update: BTO + BFO
particle/root ESS 499 / 53.3
mean lateral/speed/altitude events 2.22 / 1.72 / 1.42
mass with ≥ 2 lateral events 50.9%
seed TV max 0.760



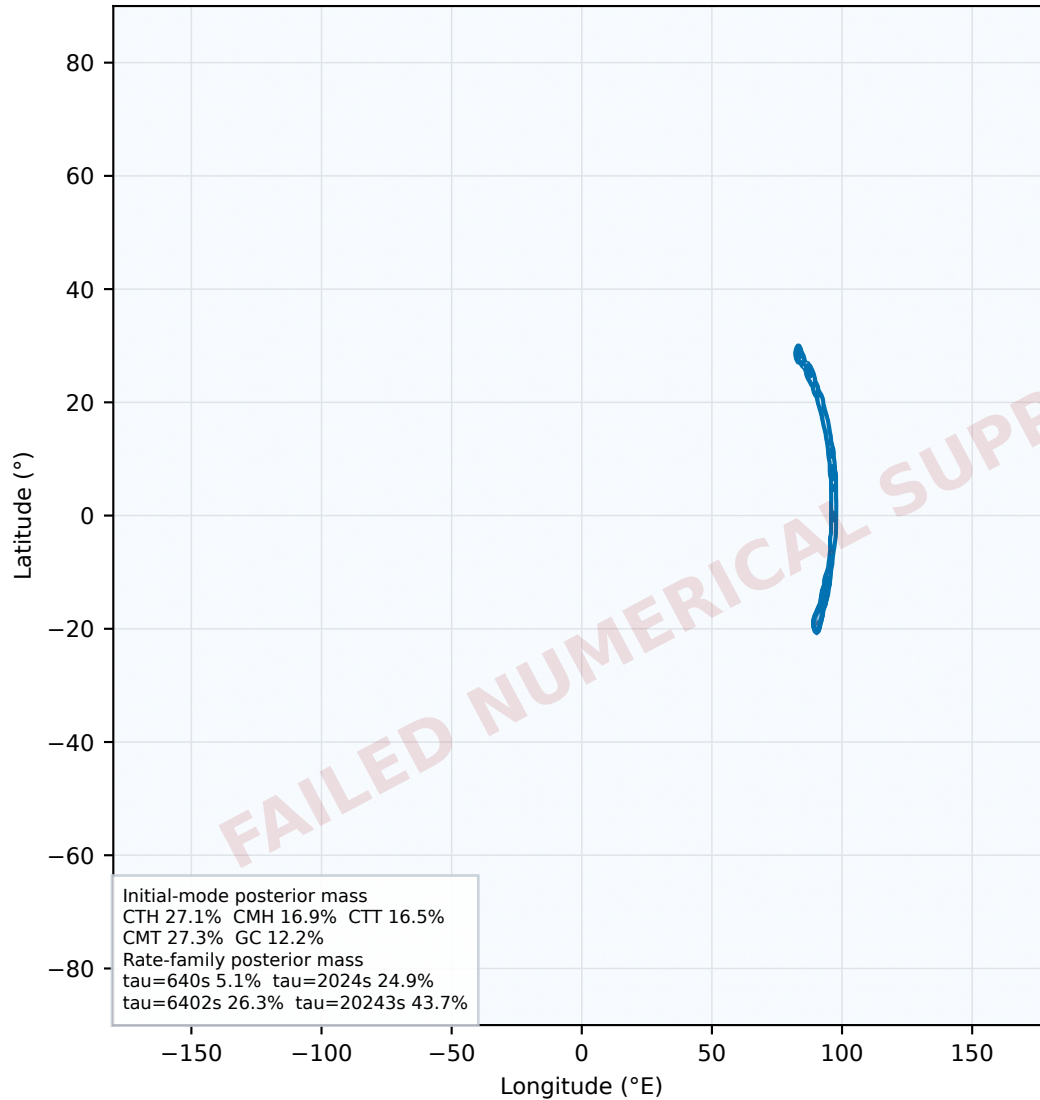
MH370 — m2041 filtering posterior



MH370 — m2141 filtering posterior

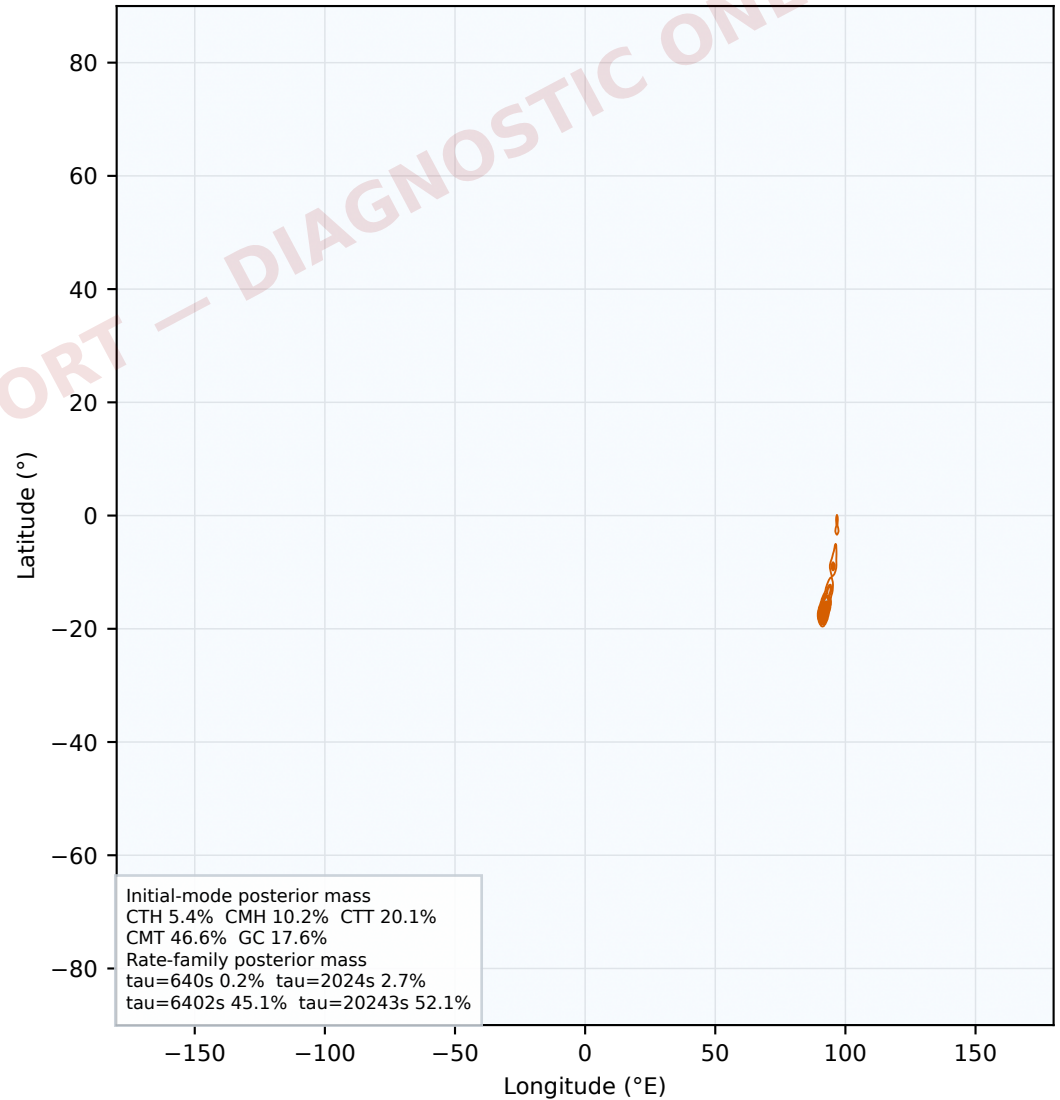
BTO only (kept separate)
SATCOM likelihood update: BTO
particle/root ESS 1355 / 22.0

mean lateral/speed/altitude events 3.95 / 3.29 / 3.36
mass with ≥ 2 lateral events 64.5%
seed TV max 0.985



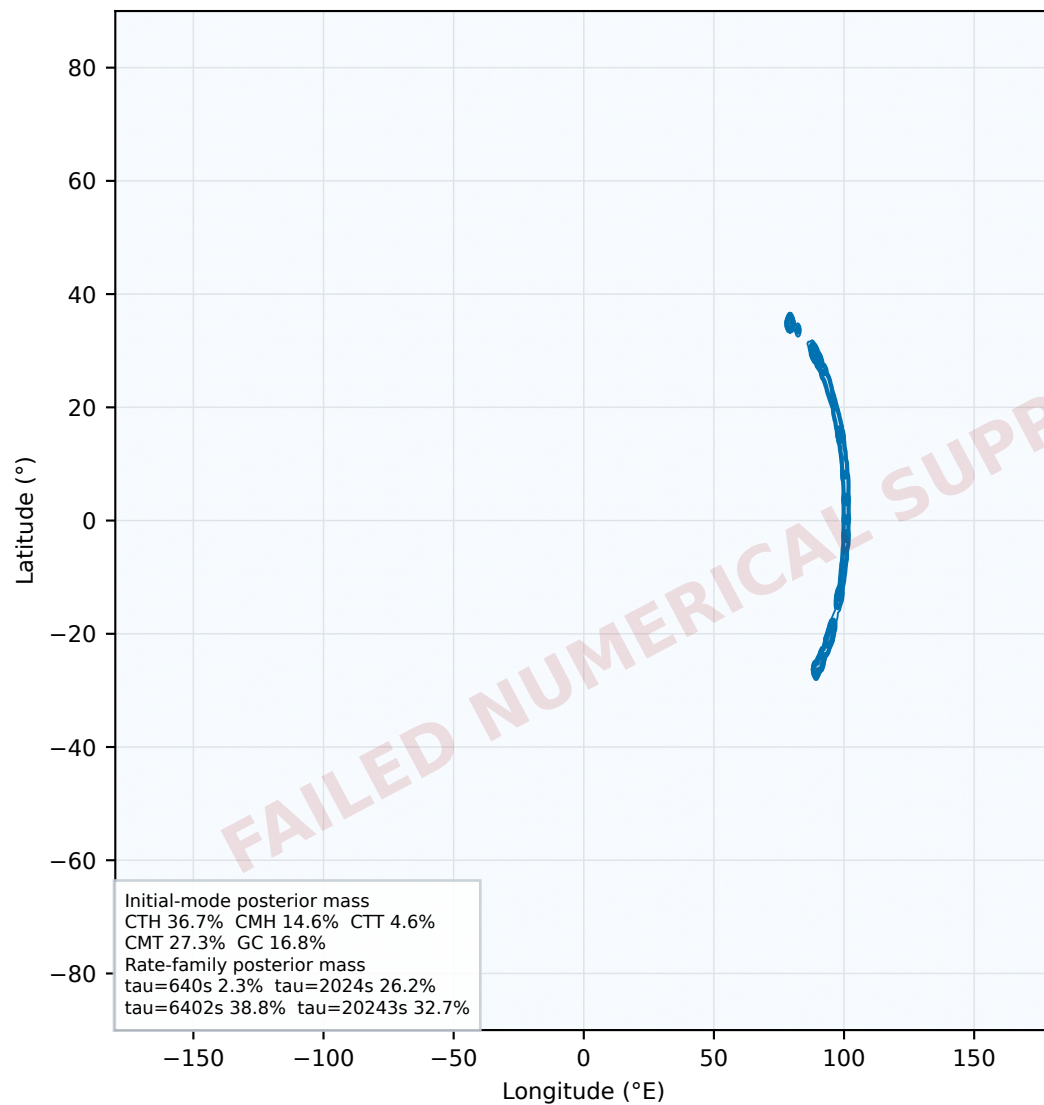
BTO + BFO (7 Hz) (kept separate)
SATCOM likelihood update: BTO + BFO
particle/root ESS 756 / 10.9

mean lateral/speed/altitude events 1.81 / 0.83 / 1.06
mass with ≥ 2 lateral events 62.6%
seed TV max 0.999

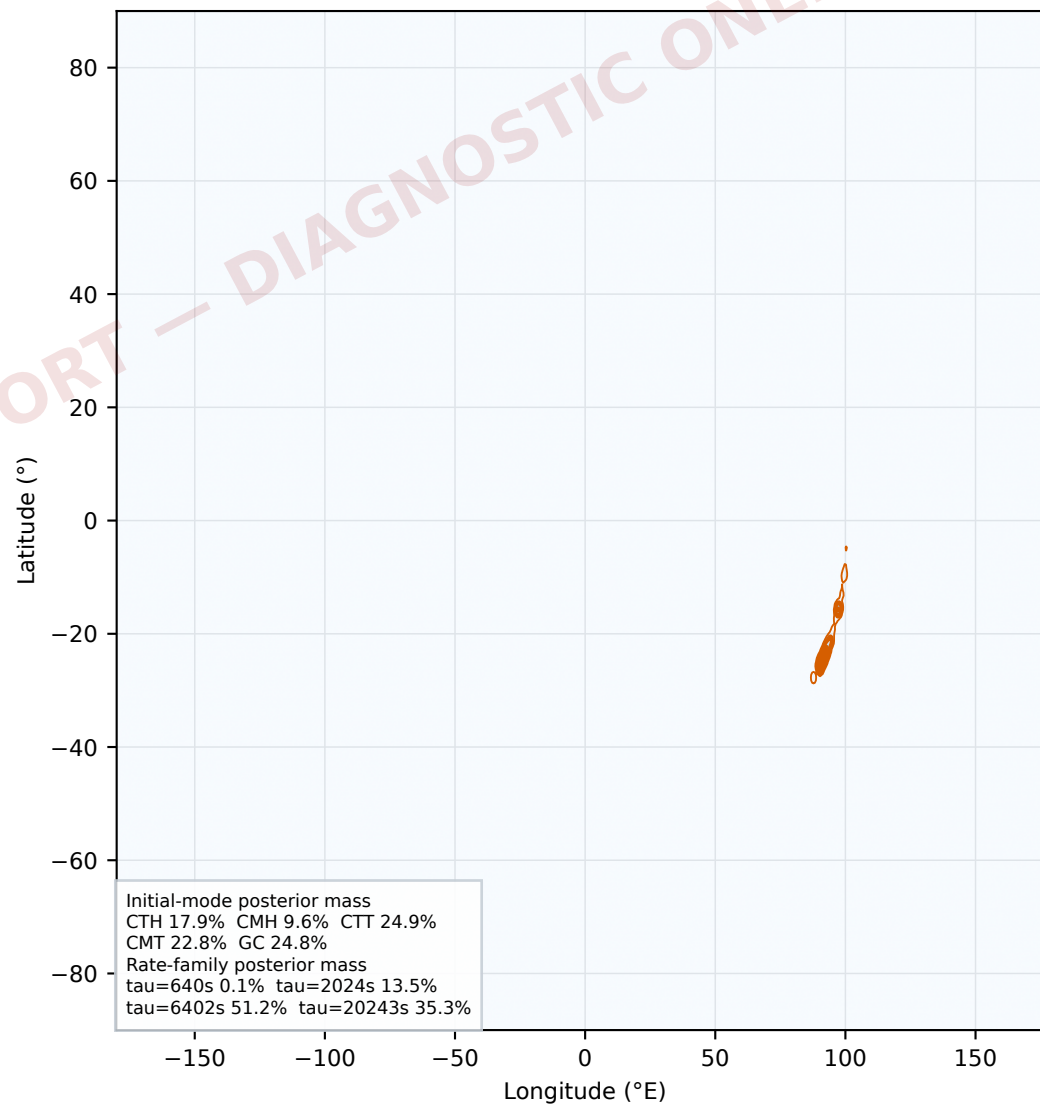


MH370 — m2241 filtering posterior

BTO only (kept separate)
SATCOM likelihood update: BTO
particle/root ESS 454 / 23.1
mean lateral/speed/altitude events 4.36 / 3.85 / 4.27
mass with ≥ 2 lateral events 88.8%
seed TV max 0.964

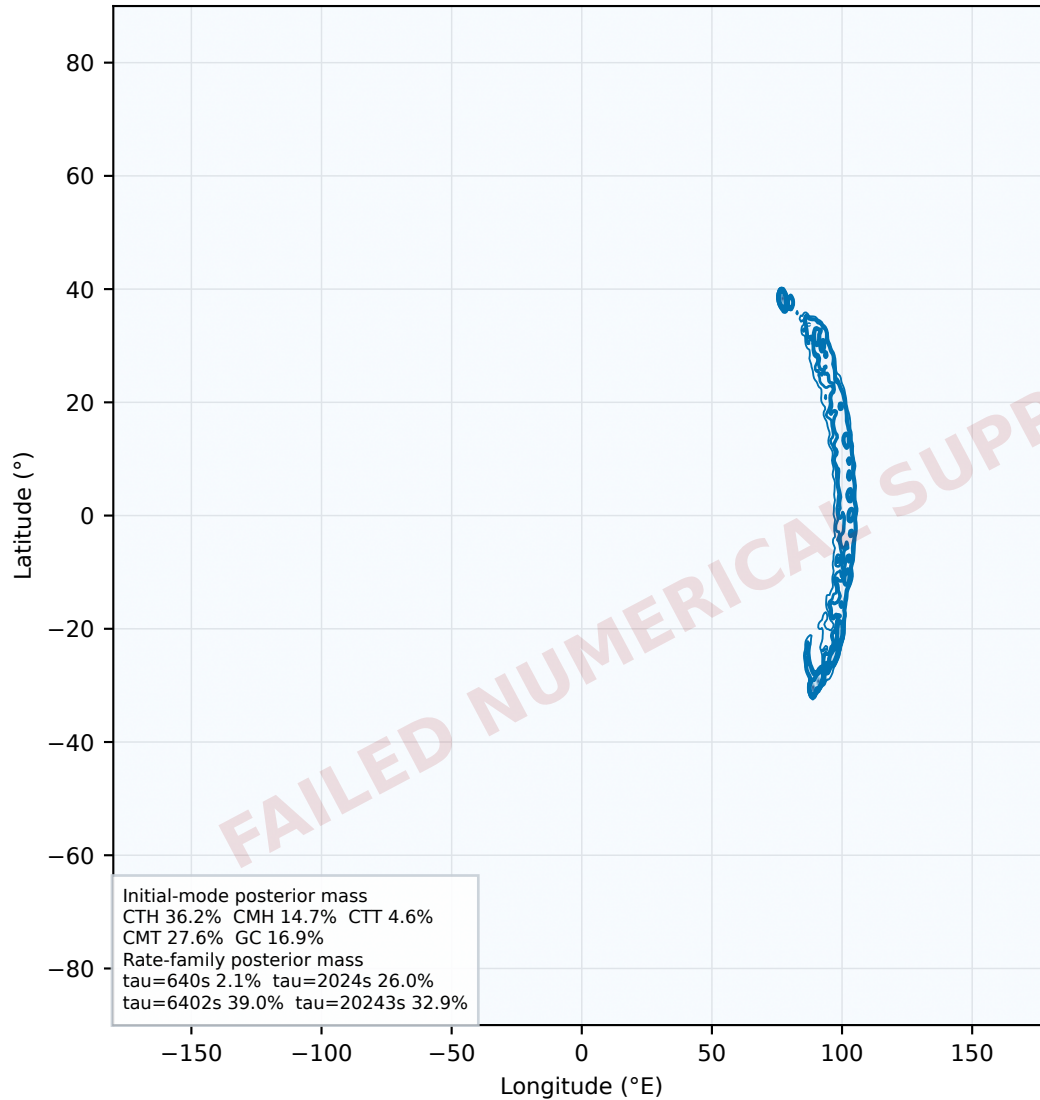


BTO + BFO (7 Hz) (kept separate)
SATCOM likelihood update: BTO + BFO
particle/root ESS 182 / 10.4
mean lateral/speed/altitude events 2.53 / 1.62 / 1.43
mass with ≥ 2 lateral events 86.7%
seed TV max 1.000

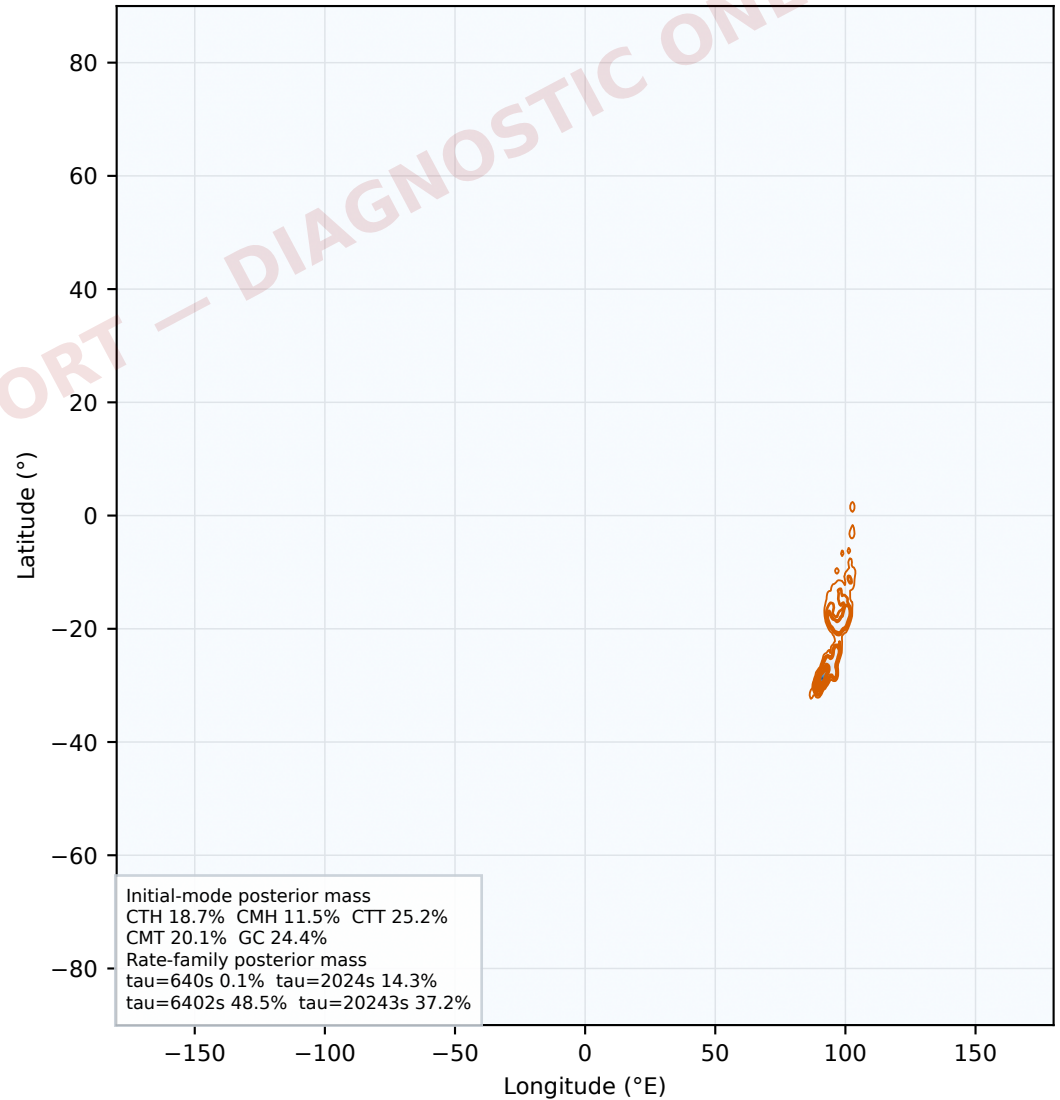


MH370 — m2315 filtering posterior

BTO only (kept separate)
propagation-only checkpoint; likelihood disabled
particle/root ESS 10115 / 22.8
mean lateral/speed/altitude events 4.74 / 4.33 / 4.69
mass with ≥ 2 lateral events 88.7%
seed TV max 0.977

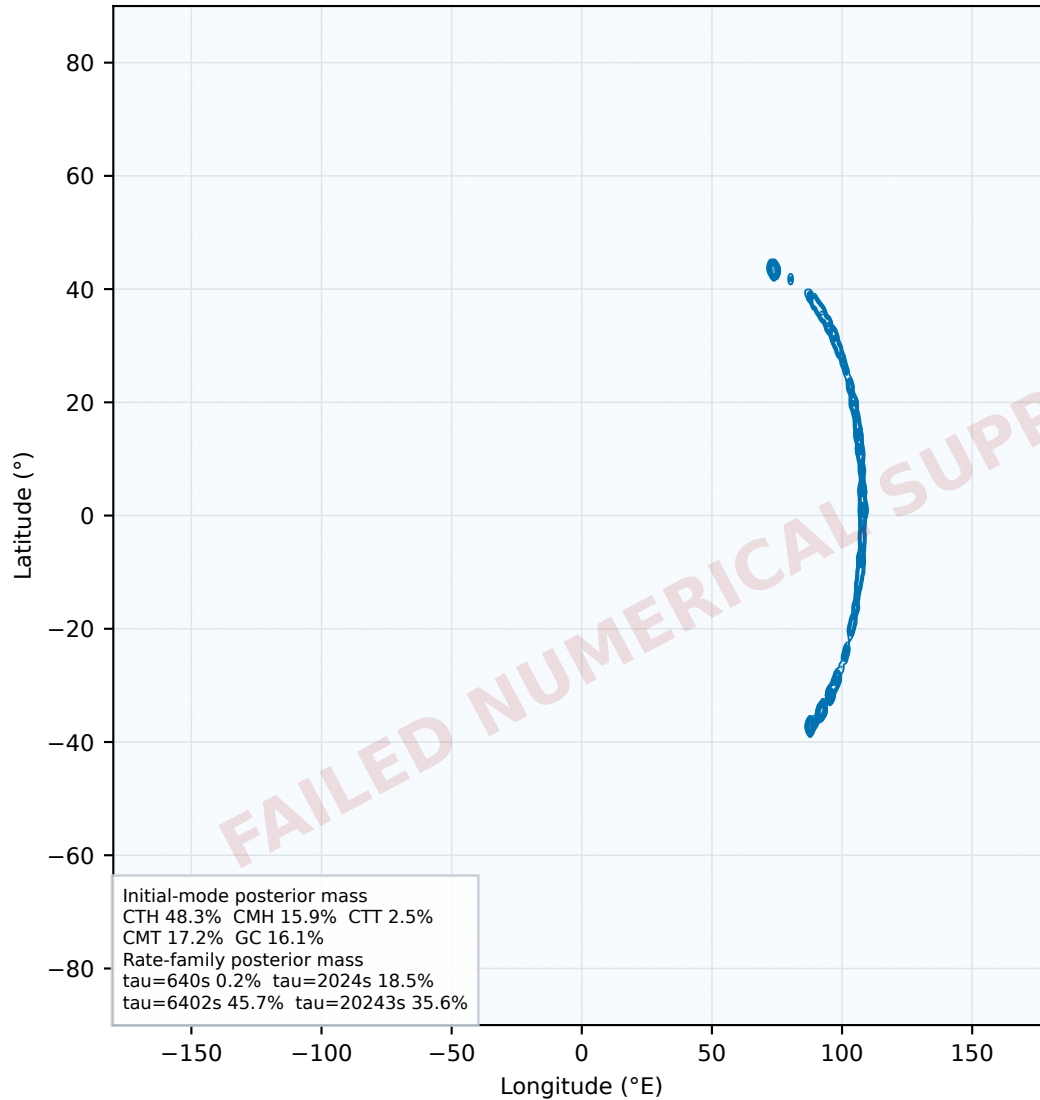


BTO + BFO (7 Hz) (kept separate)
SATCOM likelihood update: BFO
particle/root ESS 3981 / 10.6
mean lateral/speed/altitude events 2.90 / 1.97 / 1.80
mass with ≥ 2 lateral events 86.3%
seed TV max 1.000

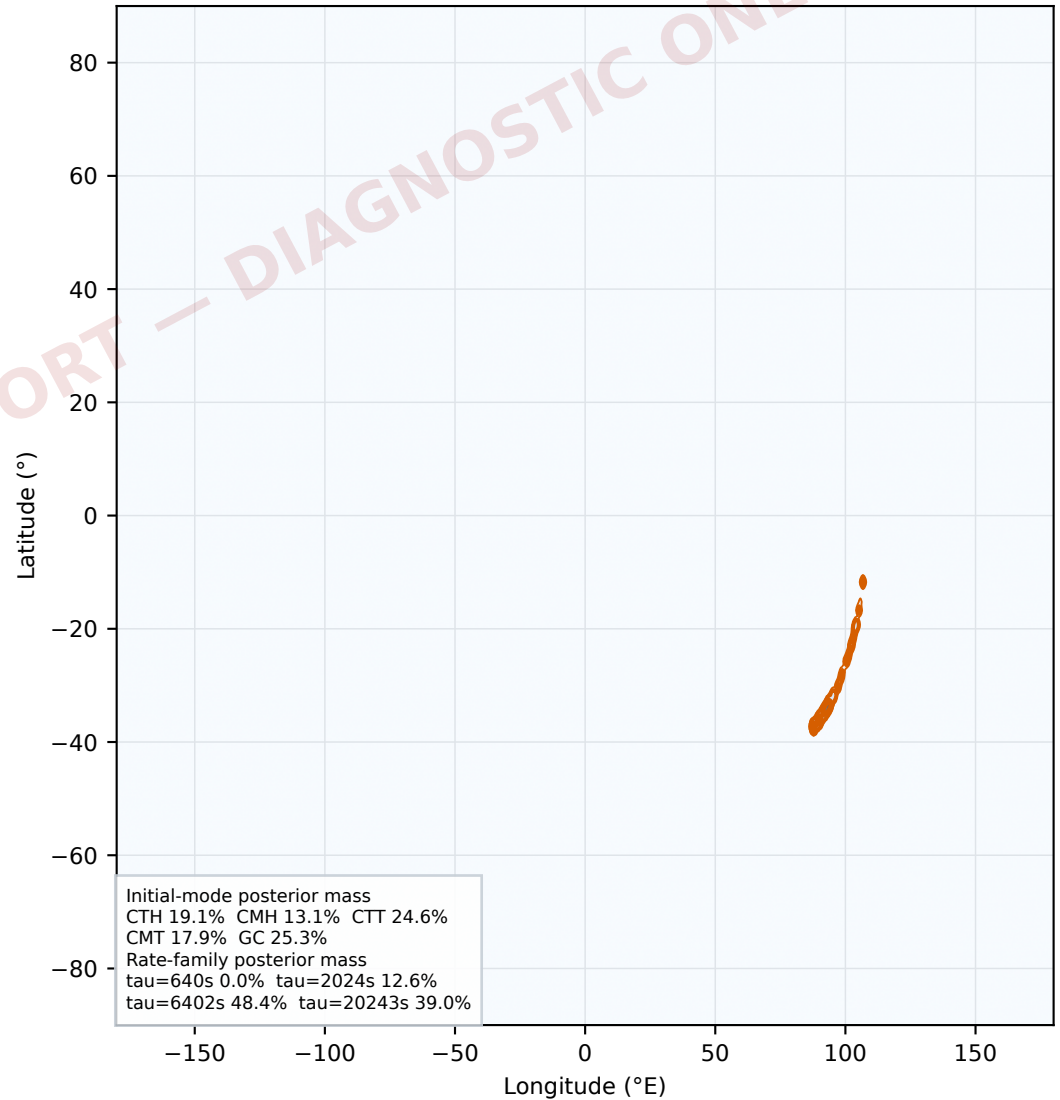


MH370 — m0011 filtering posterior

BTO only (kept separate)
SATCOM likelihood update: BTO
particle/root ESS 467 / 17.9
mean lateral/speed/altitude events 3.87 / 3.92 / 4.13
mass with ≥ 2 lateral events 79.1%
seed TV max 0.995



BTO + BFO (7 Hz) (kept separate)
SATCOM likelihood update: BTO + BFO
particle/root ESS 257 / 9.1
mean lateral/speed/altitude events 2.97 / 2.42 / 2.38
mass with ≥ 2 lateral events 87.2%
seed TV max 1.000



MH370 — structural sensitivity at m0011

BTO only; groups remain explicit conditionals inside this declared family. Differences in conditional mean location are structural spread, not Monte Carlo error.

Five declared initial lateral modes

Declared group	Posterior mass	Conditional mean lat, lon	Particle / root ESS	Mean events L / S / A	Truth error
constant magnetic heading	15.95%	10.712, 100.454	92 / 5.7	4.27 / 4.85 / 3.58	—
constant magnetic track	17.20%	-12.113, 100.052	54 / 4.5	3.95 / 3.60 / 3.07	—
constant true heading	48.26%	-1.322, 91.525	238 / 5.8	2.52 / 2.69 / 3.40	—
constant true track	2.52%	16.138, 101.261	74 / 7.4	8.49 / 9.01 / 8.16	—
great-circle continuation	16.07%	2.515, 104.311	77 / 5.7	6.71 / 6.23 / 7.36	—

Four declared recurring-event rate families

Declared group	Posterior mass	Conditional mean lat, lon	Particle / root ESS	Mean events L / S / A	Truth error
L=20243s/S=20243s/A=20243s	35.65%	-28.805, 92.265	158 / 4.8	1.56 / 1.15 / 2.02	—
L=2024s/S=2024s/A=2024s	18.51%	5.782, 106.072	128 / 9.6	8.92 / 9.61 / 9.79	—
L=6402s/S=6402s/A=6402s	45.67%	19.987, 96.410	195 / 8.1	3.53 / 3.68 / 3.37	—
L=640s/S=640s/A=640s	0.16%	10.658, 106.556	32 / 25.7	31.20 / 33.45 / 33.79	—

Masses are posterior within the named evidence family. Declared prior weights are equal and empirically uncalibrated; no cross-family mixture is formed.

MH370 — structural sensitivity at m0011

BTO + BFO (7 Hz); groups remain explicit conditionals inside this declared family. Differences in conditional mean location are structural spread, not Monte Carlo error.

Five declared initial lateral modes

Declared group	Posterior mass	Conditional mean lat, lon	Particle / root ESS	Mean events L / S / A	Truth error
constant magnetic heading	13.09%	-35.882, 90.326	319 / 1.2	2.03 / 3.03 / 0.15	—
constant magnetic track	17.93%	-34.693, 91.847	91 / 1.9	2.02 / 2.12 / 2.32	—
constant true heading	19.06%	-34.931, 91.513	46 / 1.9	3.19 / 1.19 / 3.02	—
constant true track	24.59%	-32.596, 94.379	57 / 2.1	1.90 / 3.80 / 2.05	—
great-circle continuation	25.33%	-28.964, 96.283	39 / 2.1	5.01 / 1.91 / 3.40	—

Four declared recurring-event rate families

Declared group	Posterior mass	Conditional mean lat, lon	Particle / root ESS	Mean events L / S / A	Truth error
L=20243s/S=20243s/A=20243s	38.97%	-36.033, 90.009	455 / 4.0	1.98 / 1.43 / 0.93	—
L=2024s/S=2024s/A=2024s	12.60%	-19.806, 103.607	11 / 1.0	8.13 / 3.94 / 5.59	—
L=6402s/S=6402s/A=6402s	48.43%	-33.730, 93.019	113 / 4.1	2.43 / 2.83 / 2.70	—
L=640s/S=640s/A=640s	0.00%	-3.026, 107.820	1 / 1.3	32.81 / 36.04 / 39.52	—

Masses are posterior within the named evidence family. Declared prior weights are equal and empirically uncalibrated; no cross-family mixture is formed.